Communication Systems Lecture-2: Power Spectral Density

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Energy Signals and Power Signals

Consider a **real-valued** continuous-time signal $w(t)$.

- The energy of $w(t)$ is defined as:

$$E = \lim_{T \to +\infty} \int_{-T/2}^{T/2} w^2(t) \mathrm{d}t$$

- The power of $w(t)$ is defined as:

$$P = \lim_{T \to +\infty} \frac{1}{T} \int_{-T/2}^{T/2} w^2(t) \mathrm{d}t$$

Signals can be classified as energy signals or power signals:

- Energy Signals: E is finite ($P = 0$).
  - Example: Rectangular pulse.
- Power Signals: P is finite ($E \to +\infty$).
  - Example: Sinusoidal function.

Time Average Operator

We define the time average operator $\langle . \rangle$ by:

$$\langle w(t) \rangle = \lim_{T \to +\infty} \frac{1}{T} \int_{-T/2}^{T/2} w(t) \mathrm{d}t$$

- For example, the power of $w(t)$ can be written as:

$$P = \langle w^2(t) \rangle$$

- The dc component (direct current) of $w(t)$ is defined as:

$$W_{dc} = \langle w(t) \rangle$$

  - W_{dc} is the time average of $w(t)$.

For periodic signals having a period T_0 ($w(t + nT_0) = w(t)\ \forall n \in \mathbb{Z}$), the time average operator simplifies to:

$$\langle w(t) \rangle = \frac{1}{T_0} \int_{T_0} w(t) \mathrm{d}t$$

Fourier Transform (1)

The Fourier transform of $w(t)$ is defined as:

$$W(f) = \mathcal{F}[w(t)] = \int_{-\infty}^{+\infty} w(t) e^{-j2\pi ft} \mathrm{d}t$$

$W(f)$ can be complex even if $w(t)$ is real.

$$\begin{aligned} W(f) &= X(f) + jY(f) \\ &= |W(f)| e^{j\theta(f)} \end{aligned}$$

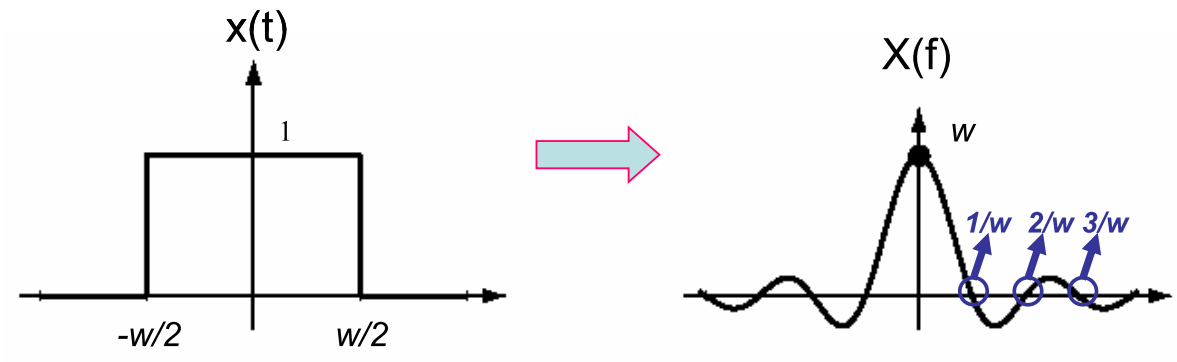
If $w(t)$ is real:

- $W(-f) = W^*(f)$.
- $|W(f)|$ is even and $\theta(f)$ is odd.
- $W(f)$ is real when $w(t)$ is even.
- $W(f)$ is pure imaginary when $w(t)$ is odd.

Fourier Transform (2)

- Fourier transform of a rectangular pulse:

$$\begin{aligned} \mathrm{rect}\,(t) &\rightarrow \mathrm{sinc}(f) \\ \mathrm{rect}\left(\frac{t}{w}\right) &\rightarrow w\mathrm{sinc}(wf) \end{aligned}$$



- Some other transforms:

$$\begin{aligned} e^{j2\pi f_0 t} &\rightarrow \delta(f - f_0) \\ \cos(2\pi f_0 t) &\rightarrow \frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0) \\ \sin(2\pi f_0 t) &\rightarrow \frac{j}{2}\delta(f + f_0) - \frac{j}{2}\delta(f - f_0) \end{aligned}$$

Power Spectral Density (1)

Consider a power signal $w(t)$.

- The power of $w(t)$ will be related to its frequency domain description by the use of a function called the Power Spectral Density (PSD).
- The PSD is a **real**, **even** and **non-negative** function of frequency denoted by $\mathcal{P}_w(f)$.
- The power in the frequency band $[f_1 \quad f_2]$ can be calculated from the PSD using the following relation:

$$P_{[f_1 \ f_2]} = \int_{f_1}^{f_2} \mathcal{P}_w(f)\mathrm{d}f$$

- The average power is the area under the PSD function:

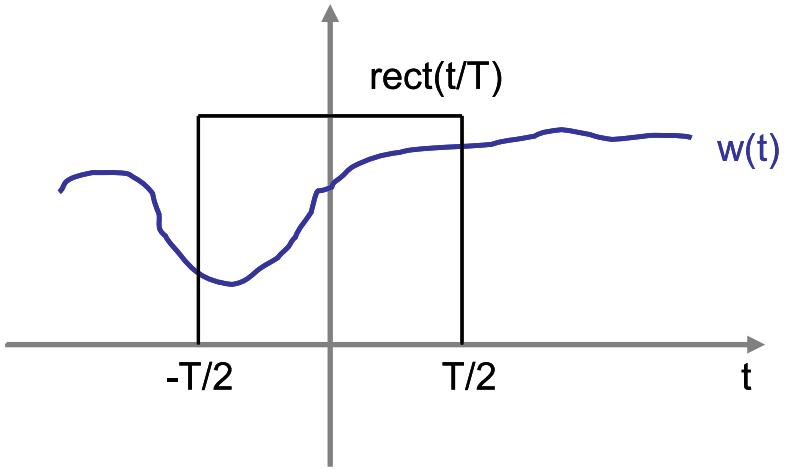
$$P = \int_{-\infty}^{+\infty} \mathcal{P}_w(f)\mathrm{d}f$$

- $\mathcal{P}_w(f)$ is expressed in W/Hz.

Power Spectral Density (2)

- To determine the PSD, we first define a truncated version of the waveform given by:

$$w_T(t) = \begin{cases} w(t), & -T/2 < t < T/2; \\ 0, & \text{elsewhere.} \end{cases}$$
$$= w(t)\mathrm{rect}\left(\frac{t}{T}\right)$$



- Evidently:

$$w(t) = \lim_{T \to +\infty} w_T(t)$$

Power Spectral Density (3)

- The average power can be written as:

$$P = \lim_{T \to +\infty} \frac{1}{T} \int_{-T/2}^{T/2} w^2(t) \mathrm{d}t = \lim_{T \to +\infty} \frac{1}{T} \int_{-\infty}^{+\infty} w_T^2(t) \mathrm{d}t$$

- Applying Parseval's theorem, the average power can be written as:

$$P = \lim_{T \to +\infty} \frac{1}{T} \int_{-\infty}^{+\infty} |W_T(f)|^2 \mathrm{d}f = \int_{-\infty}^{+\infty} \left[\lim_{T \to +\infty} \frac{1}{T} |W_T(f)|^2 \right] \mathrm{d}f$$

- On the other hand, the average power is related to the PSD by:

$$P = \int_{-\infty}^{+\infty} \mathcal{P}_w(f) \mathrm{d}f$$

Power Spectral Density (4)

- The above two equations imply that the PSD of a deterministic signal $w(t)$ can be calculated from:

$$\mathcal{P}_w(f) = \lim_{T \to +\infty} \frac{1}{T} |W_T(f)|^2$$

- From the last equation, we can directly conclude that $\mathcal{P}_w(f)$ is real, non-negative and even.

Autocorrelation function

The autocorrelation of a real power waveform $w(t)$ is defined as:

$$R_w(\tau) \triangleq \langle w(t)w(t+\tau)\rangle$$
$$= \lim_{T\to+\infty} \frac{1}{T}\int_{-T/2}^{T/2} w(t)w(t+\tau)\mathrm{d}t$$

- If $w(t)$ is periodic with period T_0:

$$R_w(\tau) = \frac{1}{T_0}\int_{T_0} w(t)w(t+\tau)\mathrm{d}t$$

From the definition of the autocorrelation function, it follows that the average power is given by:

$$P = R_w(0)$$

Wiener-Khintchine theorem: The PSD is related to the autocorrelation function by:

$$\mathcal{P}_w(f) = \mathcal{F}\left[R_w(\tau)\right]$$

Example: PSD of a sinusoid (1)

Determine the PSD of the sinusoidal waveform:

$$w(t) = A\sin(\omega_0 t)$$

- First we calculate the autocorrelation function. $w(t)$ is a periodic power waveform:

$$R_w(\tau) = \frac{1}{T_0}\int_{T_0} A^2 \sin(\omega_0 t)\sin(\omega_0(t+\tau))\mathrm{d}t$$

  - where $\omega_0 = 2\pi f_0 = 2\pi\frac{1}{T_0}$.
- Using the relation: $\sin(a)\sin(b) = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$:

$$R_w(\tau) = \frac{A^2}{2T_0}\int_{T_0}\cos(\omega_0\tau)\mathrm{d}t - \frac{A^2}{2T_0}\int_{T_0}\cos(2\omega_0 t + \omega_0\tau)\mathrm{d}t$$

  - The first integral is equal to $T_0\cos(\omega_0\tau)$ since the term $\cos(\omega_0\tau)$ is constant with respect to t.
  - Since $\cos(2\omega_0 t + \omega_0\tau)$ has a period of $T_0/2$ and since the integral of a sinusoidal function over an integer number of periods is zero, then the second integral is equal to zero.

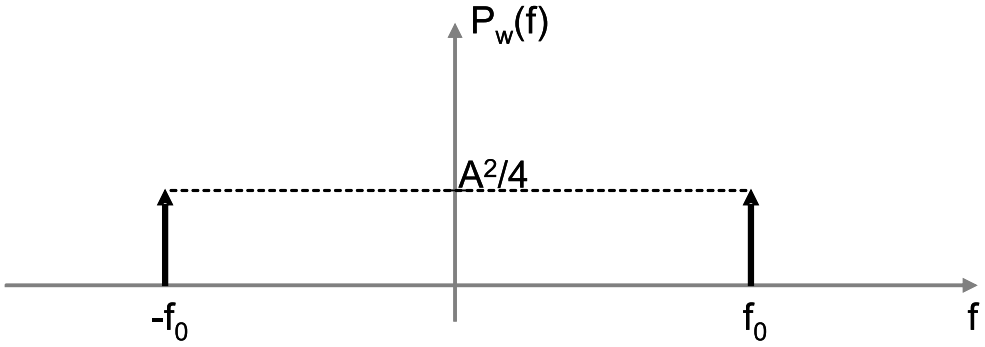
Example: PSD of a sinusoid (2)

- Consequently:

$$R_w(\tau) = \frac{A^2}{2}\cos(\omega_0\tau)$$

- The PSD can be calculated from:

$$\begin{aligned}\mathcal{P}_w(f) &= \mathcal{F}\left[\frac{A^2}{2}\cos(\omega_0\tau)\right] \\ &= \frac{A^2}{4}\delta(f+f_0) + \frac{A^2}{4}\delta(f-f_0)\end{aligned}$$



- Note that $\mathcal{P}_w(f)$ is real, non-negative and even.

Example: PSD of a sinusoid (3)

- The average power is given by:

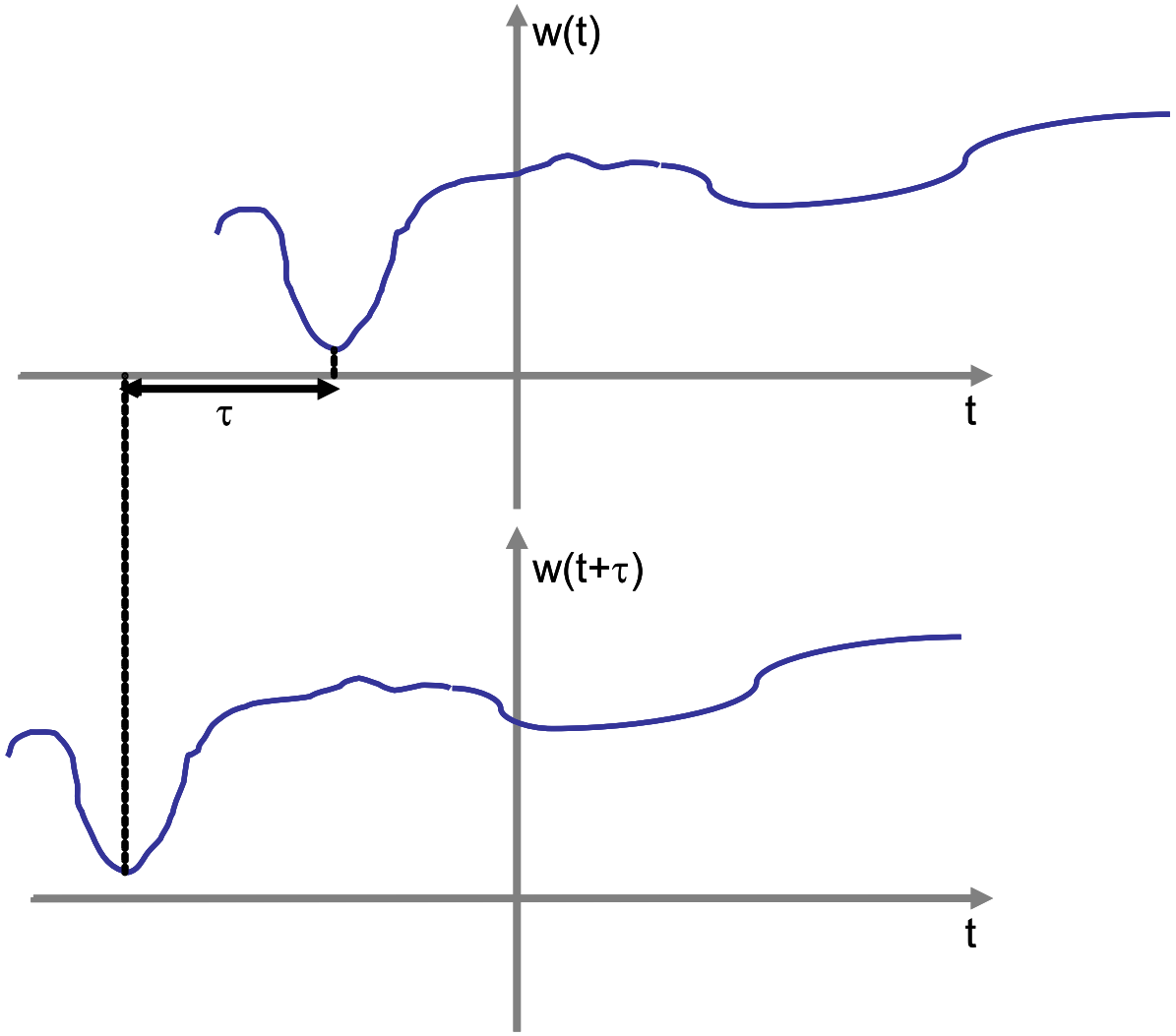
$$P = \int_{-\infty}^{+\infty} \mathcal{P}_w(f)\mathrm{d}f$$
$$= \frac{A^2}{4}\underbrace{\int_{-\infty}^{+\infty} \delta(f+f_0)\mathrm{d}f}_{=1} + \frac{A^2}{4}\underbrace{\int_{-\infty}^{+\infty} \delta(f-f_0)\mathrm{d}f}_{=1} = \frac{A^2}{2}$$

- Note also that the average power can be calculated from:

$$P = R_w(0) = \left[\frac{A^2}{2}\cos(\omega_0\tau)\right]_{\tau=0} = \frac{A^2}{2}$$

Properties of the autocorrelation function (1)

The autocorrelation function describes the resemblance of a waveform with a shifted version of itself.



Properties of the autocorrelation function (2)

The autocorrelation function is an even function:

$$R_w(\tau) = R_w(-\tau)$$

Proof:

- The autocorrelation function is given by:

$$R_w(\tau) = \langle w(t)w(t+\tau)\rangle$$

- Let $t' = t + \tau$:

$$\begin{aligned} R_w(\tau) &= \langle w(t'-\tau)w(t')\rangle \\ &= \langle w(t')w(t'-\tau)\rangle \\ &= R_w(-\tau) \end{aligned}$$

Properties of the autocorrelation function (3)

If $w(t)$ is periodic with period T_0, then $R_w(\tau)$ is also periodic with period T_0:

$$R_w(\tau + nT_0) = R_w(\tau) \quad \forall\, n \in \mathbb{Z}$$

Proof:

- $R_w(\tau + nT_0)$ can be written as:

$$R_w(\tau + nT_0) = \langle w(t)w(t + \tau + nT_0)\rangle$$

- Since $w(t)$ is periodic with period T_0, then $w(t + nT_0) = w(t)$ and $w(t + nT_0 + \tau) = w(t + \tau)$. Consequently:

$$\begin{aligned} R_w(\tau + nT_0) &= \langle w(t)w(t + \tau)\rangle \\ &= R_w(\tau) \end{aligned}$$

Properties of the autocorrelation function (4)

The autocorrelation function verifies the following relation:

$$R_w(0) \geq |R_w(\tau)| \quad \forall\, \tau$$

Proof:

- Since the time average of a nonnegative waveform is nonnegative:

$$\left\langle [w(t) \pm w(t+\tau)]^2 \right\rangle \geq 0$$

- Consequently:

$$\begin{aligned}
&\left\langle w^2(t) \pm 2w(t)w(t+\tau) + w^2(t+\tau) \right\rangle \geq 0 \\
\Rightarrow\ &\left\langle w^2(t) \right\rangle \pm 2\left\langle w(t)w(t+\tau) \right\rangle + \left\langle w^2(t+\tau) \right\rangle \geq 0 \\
\Rightarrow\ &R_w(0) \pm 2R_w(\tau) + R_w(0) \geq 0 \\
\Rightarrow\ &R_w(0) \pm R_w(\tau) \geq 0 \\
\Rightarrow\ &-R_w(0) \leq R_w(\tau) \leq R_w(0) \\
\Rightarrow\ &R_w(0) \geq |R_w(\tau)|
\end{aligned}$$